

The Refraction Hinge

A change in wave speed can produce a change in direction

IB Physics A3 worksheet

Shared structure

medium property changes → wave speed changes → wavelength changes → direction may change

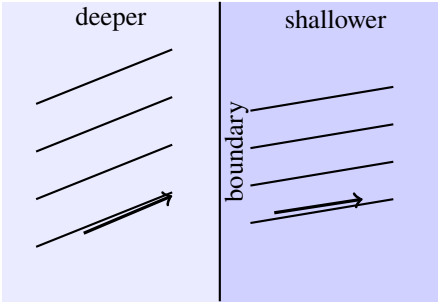
At a stationary boundary the source fixes the frequency, so

$$v = f\lambda$$

requires the wavelength to change when the speed changes. At oblique incidence, unequal slowing or speeding across the wavefront changes its direction.

1. Water waves: changing depth

A straight wavefront approaches a boundary from deeper water into shallower water at an angle.



Tasks

1. State which region gives the smaller wave speed.

2. State what happens to frequency at the boundary.

3. Use $v = f\lambda$ to predict what happens to wavelength.

4. State whether the wave bends towards or away from the normal.

5. Explain the turning using the wavefront rather than memorised ray language.

6. Explain why a wave crossing the same boundary normally changes speed but not direction.

2. Light: changing refractive index

For light in a transparent medium,

$$v = \frac{c}{n}, \quad n_1 \sin \theta_1 = n_2 \sin \theta_2.$$

Angles are measured from the normal.

A ray passes from air, $n_1 = 1.00$, into glass, $n_2 = 1.50$, at $\theta_1 = 40^\circ$.

Tasks

1. Calculate the speed of light in the glass.

2. Calculate the refracted angle θ_2 .

3. State what happens to frequency and wavelength.

4. Draw the incident ray, normal and refracted ray.

5. Explain why “greater optical density” should be interpreted quantitatively through refractive index rather than absorption.

6. Predict the direction of bending when light leaves glass and enters air.

3. Seismic waves: changing Earth layer

A seismic wave crosses from layer 1 into layer 2. Its speed changes from

$$v_1 = 6.0 \text{ km s}^{-1} \quad \text{to} \quad v_2 = 8.0 \text{ km s}^{-1}.$$

The incident angle is $\theta_1 = 30^\circ$.

For the same wave crossing a boundary,

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}.$$

Tasks

1. Calculate θ_2 .

2. State whether the wave bends towards or away from the normal.

3. State what happens to wavelength.

4. Identify two material properties that can affect seismic-wave speed.

5. Explain how refraction can make a wave path curve through a region in which speed changes gradually rather than suddenly.

6. Explain why the geometry is shared with light even though the closure determining the speed is different.

Synthesis. Complete the comparison, then state the common causal chain.

Wave	Property changing	Fixed	Changes	What sets the speed?
Water / light / seismic				

Complete: The boundary changes the _____; at oblique incidence the wavefront geometry then changes the _____. At normal incidence, speed and wavelength change but _____ does not.

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Teacher guide and answer key

Teaching focus. Students should separate the kinematic bookkeeping $v = f\lambda$ from the medium-specific relation that determines v . Refraction follows when different parts of an oblique wavefront change speed at different times. The frequency remains fixed at a stationary boundary; the wavelength changes with speed.

1. Water-wave answers

1. The shallower region gives the smaller speed.
2. Frequency remains constant because the source is unchanged and the boundary is stationary.
3. Since $v = f\lambda$, a smaller v at fixed f means a smaller λ .
4. The wave bends towards the normal.
5. The part of the wavefront entering shallow water first slows first. The other part continues faster temporarily, rotating the wavefront and therefore changing the propagation direction.
6. At normal incidence the entire wavefront reaches the boundary simultaneously. The speed and wavelength change, but there is no differential turning across the wavefront.

IB boundary. A numerical depth-speed formula is unnecessary here. The qualitative rule that shallower water gives smaller wave speed is sufficient.

2. Light answers

1.
$$v = \frac{3.00 \times 10^8}{1.50} = 2.00 \times 10^8 \text{ m s}^{-1}.$$
2.
$$1.00 \sin 40^\circ = 1.50 \sin \theta_2,$$
$$\theta_2 \approx 25.4^\circ.$$
3. Frequency remains constant. Speed and wavelength decrease.
4. The refracted ray is closer to the normal in glass.
5. Refractive index describes propagation speed through $n = c/v$. Absorption concerns loss of intensity and is a separate material property.
6. Leaving glass for air, light speeds up and bends away from the normal.

Misconception target. “Optically denser” should not be treated as meaning physically denser or more absorbing.

3. Seismic-wave answers

1.
$$\frac{\sin 30^\circ}{\sin \theta_2} = \frac{6.0}{8.0}$$
so
$$\sin \theta_2 = \frac{8.0}{6.0}(0.5) = 0.667, \quad \theta_2 \approx 41.8^\circ.$$
2. The wave bends away from the normal because it enters the faster layer.
3. Frequency remains fixed, so wavelength increases.
4. Density and elastic stiffness are acceptable.
5. A gradual speed gradient produces a sequence of very small refractions, giving a continuously curved path.
6. The geometry depends on relative wave speeds and incidence angle. The physical law determining speed differs between electromagnetic and mechanical waves.

Synthesis answers

Wave	Medium property changing	Fixed	Changes	Closure for speed
Water wave	depth	frequency	speed, wavelength, possibly direction	water depth controls wave speed qualitatively
Light	refractive index	frequency	speed, wavelength, possibly direction	$v = c/n$
Seismic wave	density and elastic properties	frequency	speed, wavelength, possibly direction	wave speed depends on stiffness and density

Model final responses

1. $v = f\lambda$ relates three wave quantities but does not identify what sets the speed in the new medium or supply the boundary geometry.
2. The boundary does not directly turn the wave; it changes *the speed, and the wavefront geometry produces the change in direction*.
3. Normal incidence, or any case in which the wave speed is unchanged across the boundary.